# PURPOSE & PATTERNS

*Simulate residential location choice and developer conversion on a parcel grid to evaluate the effect of zoning, urban-growth boundaries and transit-oriented development on urban-sprawl patterns; engine of a regional planning Digital Twin.*

## DATA INPUTS

**Parcel Cadastre** STATIC
County GIS: parcel boundaries, ownership, current use
*-> Initialize the spatial grid & developable mask*

**Census Demographics** STATIC
Tract-level household composition, income, tenure
*-> Population synthesis & preference-class assignment*

**Historical Land-Use Maps** DYNAMIC
NLCD / Landsat-derived developed-vs-undeveloped, decadal
*-> Calibration target & validation reference*

**Real-Estate Transactions** DYNAMIC
Recent sale prices, hedonic attributes, move histories
*-> Hedonic regression for amenity weights; preference clustering*

**Zoning & Policy Layers** DYNAMIC
Zoning, growth boundaries, transit corridors, road network
*-> Constrain feasible parcels & accessibility scoring*

### DATA PIPELINE

**1. Collection**
County GIS download | USGS NLCD / Landsat archive | Census microdata | MLS / assessor transaction feed

**2. Pre-processing**
Parcel-to-raster rasterization (100 m) | Land-cover reclassification (developed / undeveloped) | Spatial join of demographics to parcels

**3. Analysis**
Hedonic regression for amenity weights | k-means clustering of move patterns -> preference classes | Accessibility surface (network distance to CBD & transit)

## ABM CORE: ResLUCC (Residential Land-Use Change ABM)

### AGENTS (n=[N = households + developers])

Types: Household, Developer

**State variables:**

- income
- preference_class
- tenure_status
- current_parcel
- dissatisfaction

*-> Census Demographics + Real-Estate*

#### Submodels (Behaviors)

**Move Decision**
*Triggered by life-cycle events or dissatisfaction threshold*

**Location Choice**
*Utility-based bid over feasible parcels (multinomial logit)*

**Developer Conversion**
*Convert undeveloped -> developed if expected return > threshold*

**Amenity Update**
*Neighbor density & land cover update local aesthetic score*

### ENVIRONMENT

Grid: Regular raster of parcels
Resolution: 100 m x 100 m cells
Layers: Zoning, Accessibility, Amenity, Existing development
*-> Parcel Cadastre; Historical Land-Use Maps*

### INTERACTIONS

*** Agent <> Agent**
Households bid competitively for the same parcel; highest bid wins (auction mechanism)
Neighbor effects: nearby development alters local aesthetic & price

*** Agent <> Env**
Zoning rules constrain the feasible-parcel set
Accessibility (distance to CBD / transit) enters utility

*** Topology**
Regular 2D raster of parcels
Moore neighborhood for local effects

*-> Zoning & Policy Layers; Real-Estate Transactions*

**Temporal: 1 year | Duration: 40 years (1990-2030)**
Stop: Simulation horizon reached OR developable land exhausted

## OBSERVATIONS

**[#] Settlement Pattern Map**
Predicted developed / undeveloped raster vs observed
Spatial map
*Emergent: Leapfrog development & fragmented edges*

**[~] Density Gradient**
Built-up density vs distance to CBD over time
Curve
*Emergent: Monocentric -> polycentric structure under TOD scenarios*

## MODEL EVALUATION

### Calibration

| | |
|---|---|
| **Method:** | Pattern-oriented modeling (POM) with genetic algorithm |
| **Params:** | preference weights, developer threshold, neighbor-effect radius |
| **Data:** | Historical Land-Use Maps (1990 -> 2010 trajectory) |
| **Result:** | Best-fit parameter set; Kappa > 0.70 on calibration period |

### Validation

- Multi-resolution Kappa on held-out 2010-2020 maps
- Density-gradient & fractal-dimension comparison
- Null-model contrast (random allocation, persistence)

**Result: Pattern metrics within 10% of observed; null clearly rejected**

## SCENARIOS (if applicable):

**Baseline**
Status-quo zoning; calibrated preferences

**Growth Boundary**
Strict urban-growth boundary around existing built-up area

**TOD Upzoning**
Higher allowed density within 800 m of transit corridors

**★ Combined**
Growth boundary + TOD upzoning; key DT policy scenario